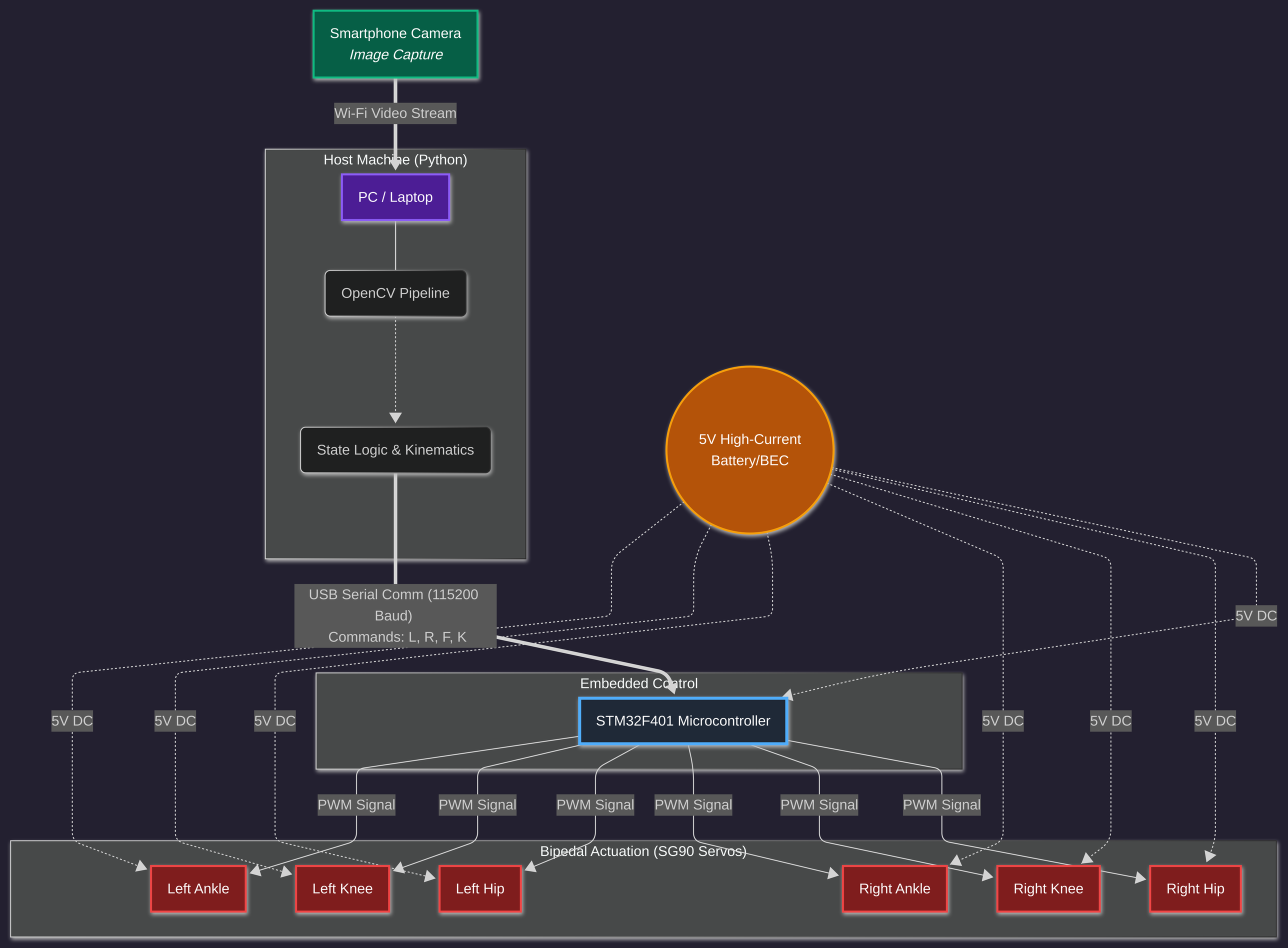
Smartphone Camera
Image Capture
Wi-Fi Video Stream
Host Machine (Python)
PC / Laptop
OpenCV Pipeline
State Logic & Kinematics
5V High-Current Battery/BEC
USB Serial Comm (115200 Baud)
Commands: L, R, F, K
5V DC
Embedded Control
STM32F401 Microcontroller
5V DC
5V DC
5V DC
5V DC
5V DC
5V DC
PWM Signal
PWM Signal
PWM Signal
PWM Signal
PWM Signal
PWM Signal
Bipedal Actuation (SG90 Servos)
Left Ankle
Left Knee
Left Hip
Right Ankle
Right Knee
Right Hip